

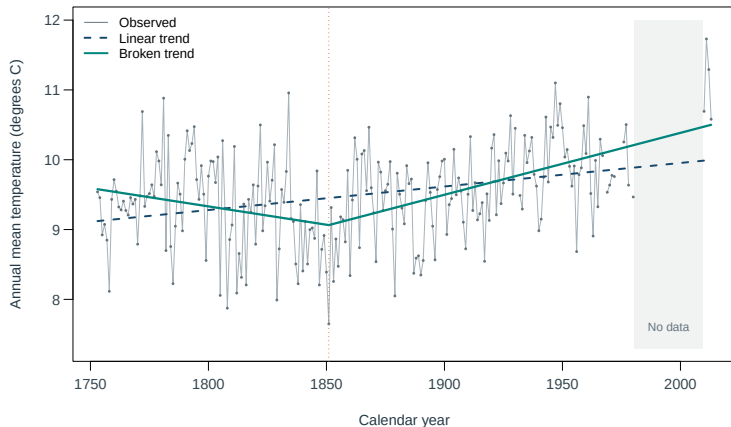
Temperature trends in Geneva

Linear trends, structural change
and bootstrap inference

Computational Methods in Econometrics
Vrije Universiteit Amsterdam
Academic year 2026–2027

Station08 1753–2013 226 observed years

Geneva's record contains a 29-year gap



226 observations
1753–2013

35 missing years
Five gaps

1981–2009
Largest gap

All observations
retained.

No interpolation.

Geneva, Switzerland. GHCN-M v4 QCF adjusted series. Annual means require at least ten valid months [2, 3].
Fitted curves across gaps are model-implied.

The linear fit leaves residual dependence

$$x_i = \frac{\text{Year}_i - 1850}{10}, \quad \hat{y}_i = 9.4481 + 0.0337x_i$$

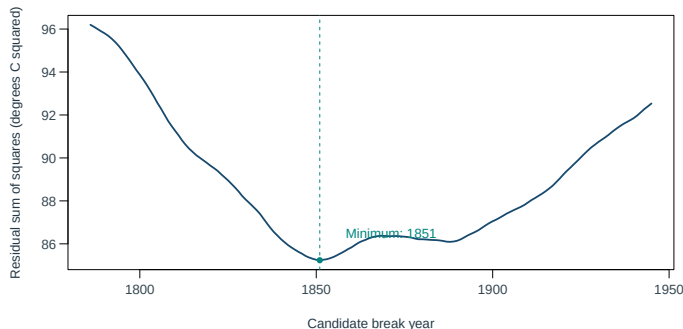
Diagnostic	Statistic	<i>p</i> -value
Durbin–Watson	1.4147	0.0002
Breusch–Pagan	0.4173	0.5183

- DW lies below the lower simulated cutoff, 1.7549.
- BP does not reject against a linear variance trend.
- Dependence qualifies conventional OLS and BP inference.

OLS slope: °C per decade. Primary DW uses observation adjacency and 9,999 iid Gaussian-null simulations. BP uses the nominal χ^2_1 reference [13, 11, 12].

The best-fitting slope change is in 1851

$$y_i = \alpha + \beta x_i + \delta(x_i - x_k)_+ + \varepsilon_i$$



15% trimming by
observation count

Candidates: 1786–1945

Before: -0.0525

After: +0.0887

°C per decade

RSS falls by 12.5%.

$F_T = RSS_0 - \min_k RSS_k = 12.2106 \text{ } ^\circ\text{C}^2$. Searching the date requires bootstrap calibration [4].

Bootstrap error generators

Method	Error draw	Preserves
IID residual	Resample residuals	Common distribution
Independent wild	Residual $\times$ independent sign	Local residual scale
Dependent wild	Residual $\times$ correlated weight	Scale and dependence

$$\text{Corr}(w_i, w_j) = \exp(-|s_i - s_j|/\ell), \quad \ell = 6 \text{ years}$$

- Generate under the fitted linear null using its residuals.
- Repeat the full date search in every bootstrap sample.

9,999 draws per test. Primary DWB bandwidth: six years. Sensitivities: three and twelve years [5, 6, 7, 8].

Break evidence depends on the bandwidth

Procedure	95% critical value	p -value	At 5%
IID residual	2.643	0.0001	Reject
Independent wild	3.426	0.0001	Reject
Dependent wild ($\ell = 6$)	10.592	0.0331	Reject
Dependent wild ($\ell = 3$)	8.772	0.0206	Reject
Dependent wild ($\ell = 12$)	12.608	0.0541	Do not reject

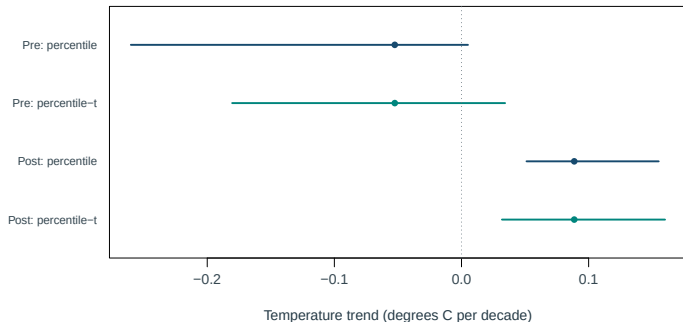
The primary DWB test rejects, but $\ell = 12$ does not.

Both independent methods reach the simulation floor.

Equal minimum p -values cannot resolve smaller tail differences.

RSS gain and critical values in $^{\circ}\text{C}^2$. Upper-tail p -value: $(1 + \text{exceedances})/(B + 1)$.

Both post-break intervals exclude zero



Pre-break

Both intervals include zero.

Post-break

Both intervals exclude zero.

Generate under the fitted broken alternative.

Reselect the date in all 4,999 draws.

DWB bandwidth six. Percentile-t intervals use calendar-time HAC studentisation [9, 10]. Interpretation is conditional on the broken-trend specification.

Eight scenarios assess size and power

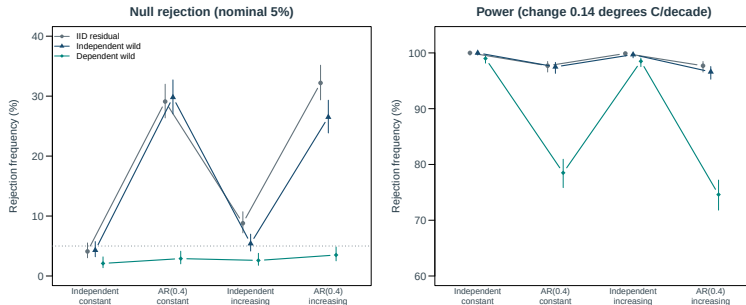
$$y_t = 9.5 - 0.05x_t + \delta(x_t - x_{113})_+ + \varepsilon_t, \quad \varepsilon_t = \rho\varepsilon_{t-1} + \sigma_t u_t$$

Factor	Values
Slope change δ	0 or 0.14 °C per decade
Dependence ρ	0 or 0.4
Innovation scale σ_t	Constant or increasing

- 1,000 datasets per scenario, 499 draws per method.
- Each dataset has 226 consecutive annual observations.
- Gaussian innovations and a central break at position 113.

8,000 datasets and 24,000 tests. Innovation scale is calibrated to 0.6182 °C. The experiment does not reproduce the observed missing-year pattern.

Independent methods over-reject under dependence



Under AR(0.4): independent methods have size 26.5–32.2%.

DWB: size 2.1–3.5% across scenarios.

95% Wilson intervals. Panel scales differ. DWB power under dependence: 74.6–78.5%.

Qualified evidence of a changing trend

- The fitted later slope is $+0.0887^{\circ}\text{C}$ per decade.
Both interval constructions exclude zero.
- The best-fitting change is near 1851.
Its test conclusion is sensitive to the dependence bandwidth.
- The simulation supports accounting for dependence.
DWB remains an approximation with finite-sample limitations.

The missing decades and model assumptions limit stronger claims.

The later slope averages 1851–2013. It is not a recent-decade rate. A single station and a fitted breakpoint cannot establish a regional transition or physical cause.

DW definitions and calibration

$$d_{\text{obs}} = \frac{\sum_{i=2}^n (e_i - e_{i-1})^2}{\sum_i e_i^2}, \quad d_{\text{cal}} = \frac{\sum_{i: s_{i+1}-s_i=1} (e_{i+1} - e_i)^2}{\sum_i e_i^2}$$

Definition	Pairs	Statistic	Lower	Upper
Observation adjacency	225	1.4147	1.7549	2.2695
Calendar adjacency	220	1.3865	1.7106	2.2211

- Common 9,999 Gaussian draws, each projected off the design.
- Type-6 cutoffs are order statistics 250 and 9,750.
- Strict rejection outside cutoffs agrees with $p \leq 0.05$.

Both two-sided p-values are 0.0002. Exact rank size is conditional on the continuous iid Gaussian null and fixed design [13].

Testing and interval bootstraps target different models

Break test

Null mean $\hat{y}_0$ and residuals e_0 .

Scale residuals by $\sqrt{n/(n-2)}$.

Repeat the full search and count $F_T^* \geq F_T$.

Slope intervals

Alternative mean $\hat{y}_1$ and residuals e_1 .

Scale residuals by $\sqrt{n/(n-3)}$.

Reselect the date and recompute slopes and standard errors.

$$J_i = [1, x_i, (x_i - \tau)_+, -\hat{\delta} 1(x_i > \tau)], \quad K_{ij} = \max\{1 - |s_i - s_j|/7, 0\}$$

Studentisation includes the local date direction and uses correction $n/(n-4)$. A rescaled indicator replaces the fourth column for numerical stability when the kink is nonzero. Weak identification can limit coverage.

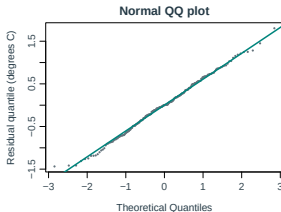
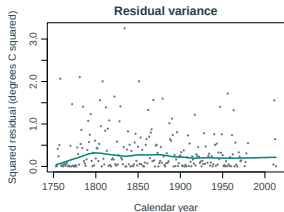
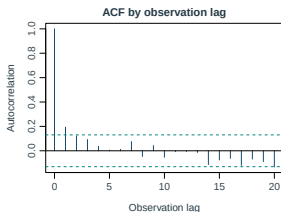
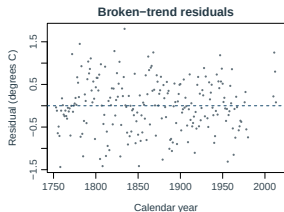
The main shape persists in reduced samples

Sample	n	Break year	Pre slope	Post slope
Full sample	226	1851	-0.0525	0.0887
Omit 1851	225	1853	-0.0464	0.0872
End in 1980	222	1850	-0.0477	0.0776

- Each reduced sample repeats the trimmed break search.
- These are descriptive checks without new bootstrap tests.
- Primary inference retains all 226 observations.

Slopes are °C per decade. Alternative-bootstrap date percentiles are 1793 and 1896, with 1.54% of selections at boundaries. These do not constitute a calibrated break-date confidence set.

Dependence remains after fitting the break



ACF(1):
0.290 to 0.194

DW:
1.415 to 1.613

RSS reduction:
12.53%

Dependence remains.

Observation lags in the ACF. Selecting a date changes the design, so the linear-model DW critical values do not calibrate a test here.

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